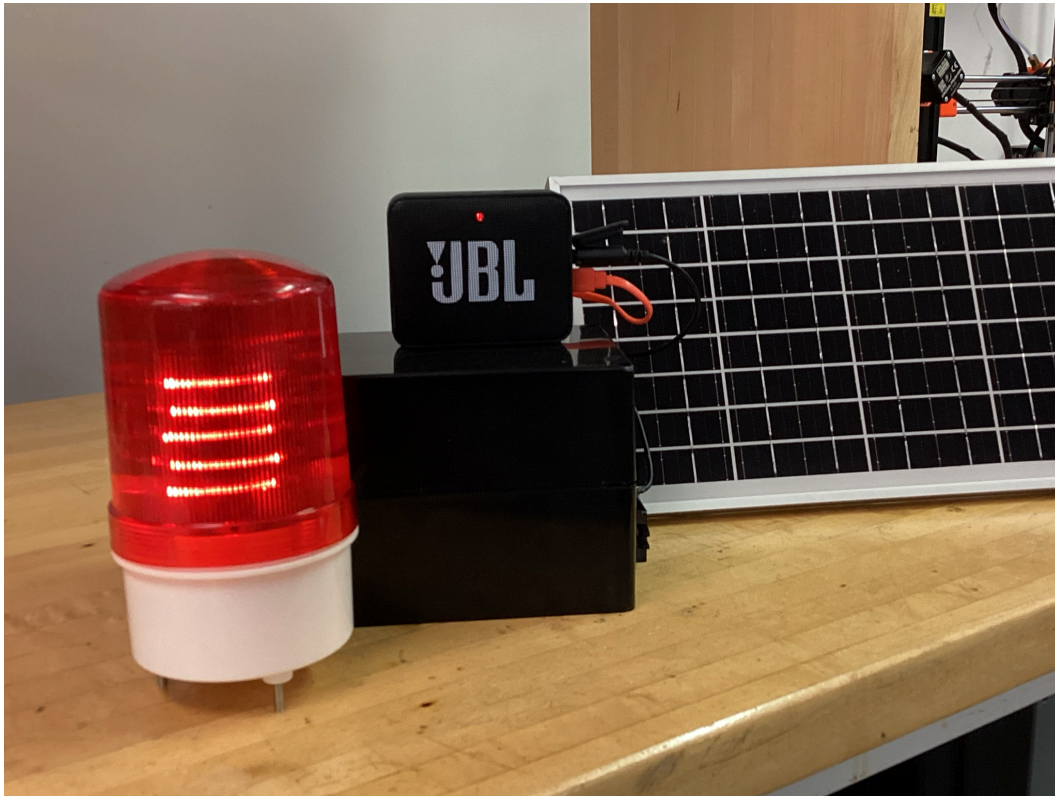


Slow Your Roll



Prepared for

Jon Kuester and Blake Lanphier
Historic Wagner Farm

Submitted by

Tali Smith, Mika Kato, Audrey Toman, Victoria Sefan
Team 3, Section 7

Sponsored by

Design Thinking Communication Program
Professor Stacy Benjamin and Professor Shuwen Li
McCormick School of Engineering
Northwestern University
Evanston, Illinois 60208

Date

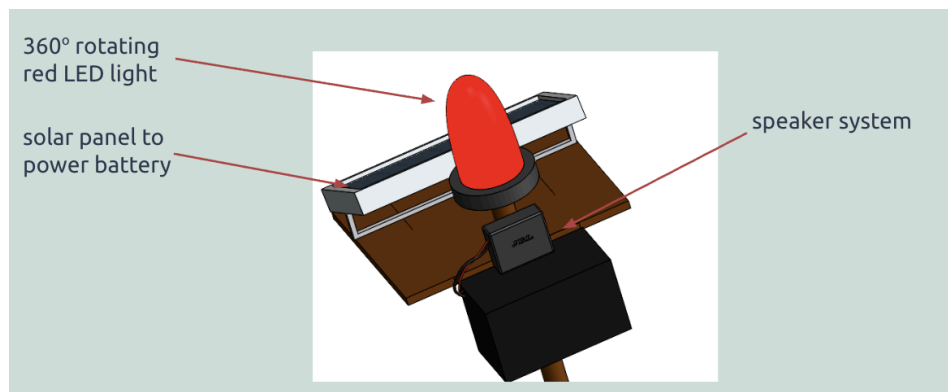
6/9/2025

Executive Summary

Historic Wagner Farm – a living history museum and dairy farm – has been working to address persistent challenges posed by migratory birds, particularly geese that interfere with pasture-raised cattle and chickens. While the surface-level need is to reduce geese presence in pasture areas, the deeper challenge lies in mitigating the risk of Avian Influenza transmission, protecting both animal health and the sustainability of pasture-based farming. Current deterrent methods are often labor-intensive, ineffective, or disruptive to farm operations.

To resolve the problem, we interviewed two staff members at Historic Wagner Farm to understand the specific challenges posed by geese, and conducted multiple on-site visits to observe pasture conditions, animal movement, and existing deterrent efforts. Through developing and testing four mockups focused on power supply, weather resistance, and deterrent effectiveness, we developed a prototype that meets the farm's needs.

Our solution—Slow Your Roll—is a solar-powered, post-mounted deterrent system designed to reduce goose presence without disturbing livestock. It is a standalone unit that activates every 20-30 minutes using a programmable timer, eliminating the need for motion sensors that could be triggered by cows. The system, shown in Figure 1, consists of a 10W solar panel, a strobe light, and a speaker that plays pre-recorded goose deterrent sounds. These components are mounted on a durable wooden post, with a weatherproof electrical enclosure protecting the battery, MP3 module, and relay timer. Solar energy allows for day/night operation with minimal maintenance.



Slow Your Roll offers Wagner Farm a reliable, low-maintenance, and eco-friendly solution to a persistent and costly problem. Further testing in varied weather conditions is recommended, along with exploring more varied deterrent sounds and strobe patterns to prevent geese habituation. Integrating a battery level indicator could also improve long-term usability.

Table of Contents

Table of Contents.....	3
List of Figures.....	4
List of Tables.....	5
I. Introduction.....	6
II. Users and Requirements.....	7
III. Slow Your Roll Rationale.....	8
IV. Next Steps.....	13
V. Conclusion.....	14
Appendix A Secondary research summary.....	15
Appendix B Summary of Observations at Wagner Farm and Interviews with project partners and biologist.....	19
Appendix C Project Definition.....	25
Appendix D Mockup testing and feedback summary.....	28
Appendix E Instructions for construction and bill of materials.....	32
Appendix F Instructions for use.....	41

List of Figures

Fig. 1. Final Design	7
Fig. 2. Labeled parts of Slow Your Roll	9
Fig. 3. Creek in Cow Pasture of Historic Wagner Farm	9
Fig. 4. 360° Rotating Siren Light place atop the pole	10
Fig. 5. JBL Speaker place on the side the pole	11
Fig. 6. Timer System	12
Fig. 7. System Mounted	13
Fig. A1. A potential decoy predator	17
Fig. B1. A photo of the general area within Wagner Farm.	20
Fig F.1. Wind Chimes Image	29

List of Tables

Table 1. Project Requirements	26
Table 2. Bill of Materials used to construct Slow Your Roll	33

I. Introduction

Canada geese are a familiar sight across the Midwest, but their growing presence on farms presents serious health and operational challenges. For Wagner Farm, in Glenview, Illinois, the geese are a leading spreader of avian influenza – a dangerous virus spread through feces that poses a threat to both chickens and cattle. Despite attempts to control the issue through hired contractors and deterrent devices, previous solutions have proven to be either too expensive, too disruptive, or too labor-intensive to sustain (see Appendix A for an overview of common deterrent tools). Our design, Slow Your Roll, aims to address this gap by providing a durable, affordable, and humane goose deterrent system that can operate continuously with minimal maintenance.

The staff at Wagner Farm, the primary users, have noted that geese gather most frequently in the cow pasture, particularly around a small creek. Their concern is not just the mess the geese leave behind, but the serious risks their presence creates. The farm's chickens, though enclosed in a mobile coop, risk exposure to contaminated ground as the coop is moved. Likewise, cows grazing in the pasture could ingest infected materials, potentially leading to health complications and disruptions in milk production (see Appendix B for details on geese activity zones across the farm). Current deterrents – like solar lights and laser systems – have helped temporarily but either require constant oversight, cause unwanted side effects (e.g., disturbing the neighbors or livestock), or are too costly to maintain long-term. In particular, the farm has tried hiring an external goose-chasing service that used dogs and laser systems, but these options were extremely expensive and only effective during the day, leaving the farm vulnerable overnight.

Through interviews and site visits, our team learned that an ideal solution must operate autonomously, work during nighttime hours when geese often land, and avoid interfering with the daily rhythms of farm life (Appendix B). The staff emphasized that the system must not harm the geese, must comply with suburban noise and light ordinances, and must remain safe for livestock and visitors. These constraints led us to explore a multi-sensory deterrent system that combines unpredictable light and sound with flexible positioning and solar power.

Our solution—Slow Your Roll (Fig. 1)—is a solar-powered deterrent system featuring a 360° rotating siren light and an integrated speaker that emits goose distress calls. Designed to be mounted securely into the ground near the creek, the device uses a rechargeable battery to function overnight and can be relocated to prevent geese from becoming habituated to its presence. With minimal user input required, the system aims to deter geese from nesting or gathering, thereby reducing the risk of avian influenza transmission.



Fig. 1. Final Design

By creating a low-maintenance, livestock-safe solution, our goal is to support the farm in protecting its animals and maintaining a clean, educational space for the community. This report details the user needs and environmental constraints at Wagner Farm, the rationale and development of our Slow Your Roll prototype, and outlines future steps for improving the design through enhanced durability, weatherproofing, and motion-activated capabilities.

II. Users and Requirements

To better understand the needs of Historic Wagner Farm and identify the main problem areas, our team visited the farm to conduct an initial interview with Jon Kuester, the farm's director (see Appendix B). During our visit, Jon walked us around the grounds and told us how each area was impacted by the geese, including the cow pasture, chicken coop, and public walkways. He also explained more about the farm's past experiences with deterrent methods and emphasized the importance of keeping the geese away from the chickens and livestock to protect animal health and maintain a clean and safe environment for visitors.

Users

Primary User. The primary users are the staff of Historic Wagner Farm. They are responsible for setting up and maintaining the geese deterrent systems. The staff will mainly interact with the

design in the cow pasture in order to install and repair it in case of damage from harsh weather conditions or livestock.

Secondary Users. The secondary stakeholders include both the livestock and the visitors of Wagner Farm. The livestock are directly affected by the presence of the geese, as geese can transmit avian influenza, posing a serious threat to animal health. Additionally, Wagner Farm has many visitors, a lot of whom are children, so minimizing health risks is essential to maintaining a safe and educational environment.

Major Requirements

Jon and Blake mentioned multiple key requirements in which Slow Your Roll had to meet to be an effective solution (see Appendix C for main requirements).

Cost-Efficiency: A main issue that arose with previously used deterrents was that each of the solutions that they tried, such as dogs to chase away the geese or lasers (see Appendix B), cost a lot of money (about \$1200 a month). These services also required human supervision and were limited to daytime operation, which left pasture areas unprotected overnight. Therefore, the farm specified that they wanted the device to be relatively cost effective and capable of continuous operation.

Minimal maintenance: The laser system previously used by Wagner Farm (see Appendix B) was not an efficient solution as it required someone to constantly be with the laser making sure to point it where the geese were. Therefore, the design needs to be easy to maintain, requiring little maintenance and setup.

Durability: As this design is going to be outside 24/7 it is vital that the device will be able to withstand all sorts of weather conditions, such as rain, wind, and varying temperatures.

Unpredictability: Geese quickly adapt to static or repetitive deterrents, so the device must be unpredictable in order to remain effective over time.

III. Slow Your Roll Rationale

Slow Your Roll minimizes ongoing maintenance and can be relocated across pasture zones as needed through the use of solar power and a sealed battery system. The device is designed to be a cost-effective, 24/7 deterrent system tailored to Wagner Farm's specific needs. It combines light and sound deterrents that activate on a timer system (Fig 2). The goal is to deter geese from settling or nesting overnight in high-risk pasture areas without requiring staff presence.

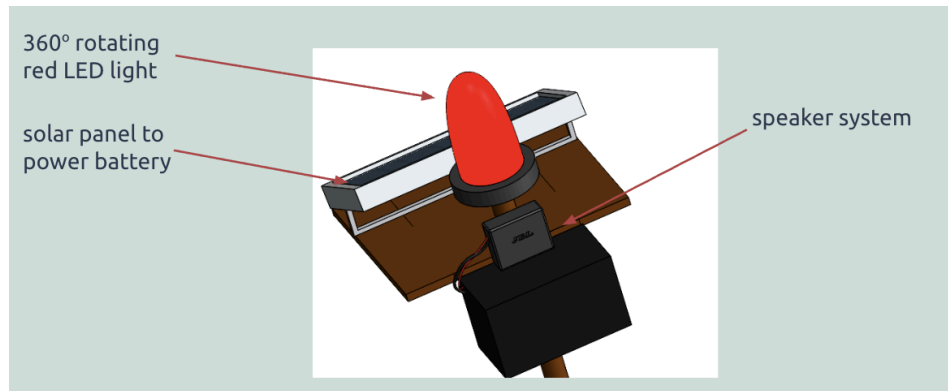


Fig. 2. Labeled parts of Slow Your Roll

Slow Your Roll is an auditory and visual deterrent system aimed at deterring geese from the creek located in the cow pasture of Wagner Farm (Fig. 3). During intervals throughout the night the battery will turn on powering a 360° light system and speaker. In addition the battery will be set on a randomized timer so the light and sound will run on unpredictable cycles ensuring that the geese do not get used to the light system, making it a more long-lasting solution. Using multiple units increases effectiveness, emitting sound and spreading light across a larger area.



Fig. 3. Creek in Cow Pasture of Historic Wagner Farm

360 Rotating Siren Light

The 360° rotating red light serves as a visual deterrent for the geese. During our farm visit, Jon said that red light is seen by the geese but not the cows so the light will not be a disturbance to

them. During our on-campus testing (see Appendix D), we learned that a static light would not be effective enough to scare away the geese. For this reason the light will be constantly rotating. This will not only cover the greatest amount of surface area, but the frequent shining of light in the geese's eyes will serve as an annoyance to the geese. The light will be placed at the top of the post to increase visibility of the light and allow geese flying over to be able to see it.



Fig. 4. 360° Rotating Siren Light place atop the pole

Speaker System

The speaker system serves as an auditory deterrent for the geese. The system features a JBL speaker which emits a 12V MP3 sound module, which is attached to the side of the pole and powered by the battery. From our on-campus and farm testing we noticed that the geese were not scared by the clinging metal of our wind chime mockup (see Appendix D); therefore, the design uses harsher, more unpredictable noises. The speaker will play a mixture of geese distress calls and predator noises. Our interview with a biologist at Lincoln Park Zoo supported this approach, noting that unfamiliar predator calls are more effective in urban-adapted geese than typical farm noises(see Appendix B). By mixing the two, the geese will not get accustomed to the noise. If the geese think it is unsafe to land they will avoid congregating in the area. The speaker also runs on the randomized timer to increase unpredictableness.



Fig. 5. JBL Speaker place on the side the pole

Solar Panel

The solar panel charges the 12V battery that powers the light and speaker. It is a 10W monocrystalline solar panel, with an adjustable base. The solar panel is IP65 certified and extremely durable being able to withstand wind pressure up to 2400Pa and 5400Pa and would work well in -40°F-185°F. The solar panel is attached to the side of the post.

From our site visit we noticed that the creek is quite far away from the barn area, making it more time-consuming for staff members to access. Therefore, a solar panel minimizes the need to change the battery since it is charged by the sun during the day.

Timer System

A programmable timer relay is connected between the battery and the sound system, turning the device on for 15-30 second bursts every 20–30 minutes throughout the night. This randomization of the light and sound will create an unpredictability of the design that will prevent the adaptation of the geese to the design allowing it to maintain long-term effectiveness. This system also allows the battery to last longer as it will not be running constantly throughout the night.

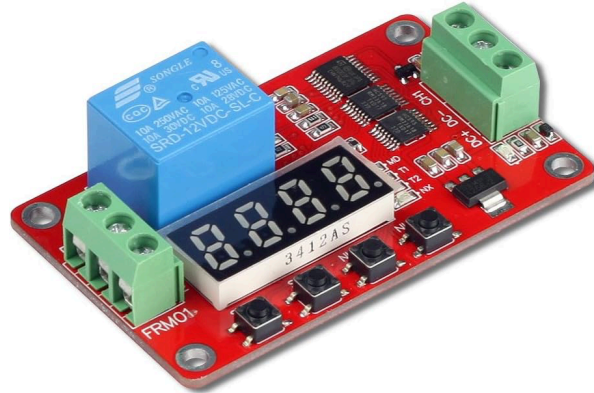


Fig. 6. Timer System

The battery, timer system, and MP3 module will be housed in a 7.87in x 4.72in x 4.45in IP65 waterproof junction box.

Battery System

Slow Your Roll is powered by a 12V sealed lead acid battery housed within a weatherproof junction box mounted on the post. This battery type has low maintenance needs and it is compatible with solar charging. It connects to the timer system and MP3 module, enabling reliable activation of both the light and sound components. The solar panel charges the battery during the day, and a built-in charge controller prevents electrical damage by managing voltage flow.

Randomization System

To maintain long-term effectiveness, the system includes a programmable relay timer that controls when the light and sound activate. The time is set to trigger 15-30 second bursts at randomized intervals throughout the night, typically every 20-30 minutes. This irregular pattern prevents the geese from becoming habituated to the stimulus, which is critical for sustained deterrence.

Mounting System

The entire system is mounted on a vertical PVC pipe post with an integrated ground anchor for stability (Fig 7). The solar panel is positioned at the top to maximize sun exposure, while the red siren light is mounted directly above the post to maximize visibility to geese overhead. Midway down, a waterproof enclosure box houses the battery and electronics, and the speaker is affixed to the side of the post for optimal sound projection (see Appendix F for detailed instructions for placement and installation).



Fig. 7. System Mounted

IV. Next Steps

Slow Your Roll is a functional deterrent system that will be delivered to Wagner Farm for testing of the various components implemented. Below are the factors that should be evaluated.

Recording effectiveness

To optimize Slow Your Roll's use, it is crucial that geese populations are tracked over time. Due to price constraints, a camera could not be installed on our prototype; however, a camera can easily be strapped on the mount to track geese population at nighttime to track Slow Your Roll's effectiveness.

Battery Life Testing

In addition, while the product is designed to have a battery for reserve electricity, future improvements would involve increasing the power stored in the battery in case of consistently bad weather. Wagner Farm should test the unit's battery life across multiple cloud days, ensure reliable solar charging, and confirm the system continues operating under wet, wind, and cold conditions.

Durability testing

Our design has yet to be tested on Wagner Farms concerning its ability to stay balanced in the dirt. In the future Wagner Farm should test the design's ability to stay steady in varying weather conditions.

V. Conclusion

To summarize, Slow Your Roll meets the conditions required to deter geese on the cow pasture in Wagner Farm, which will aid in preventing the spread of avian influenza to the livestock. This design contains two key features that will deter the geese: A rotating red light and a speaker that emits predator and goose distress calls at random increments. These components are powered by a solar panel that is connected to a battery box, and all of the components will be mounted on a wood pole near the creek area in the cow pasture.

Slow Your Roll allows Wagner Farm to reduce the spread of avian influenza on the farm by targeting a location where the geese are the most prevalent (the creek) at a time when external disruptions are least likely to deter geese from the farm (during nighttime).

Appendix A Secondary research summary

Introduction

Avian influenza is a serious viral disease that has caused widespread outbreaks in the United States in recent years. It spreads rapidly through wild and domestic migratory bird populations – especially geese and ducks – who transmit the disease through their droppings. This contamination threatens not only commercial poultry operations but also smaller farms, prompting widespread adoption of deterrents and biosecurity protocols to protect livestock, workers, and the public. Historic Wagner Farm encounters a similar challenge. As a publicly owned farm with pasture-raised chickens and cattle, Wagner Farm faces an elevated risk of avian influenza transmission due to frequent goose activity in its open pasture areas. While the farm has implemented basic preventative measures, geese continue to gather in key areas, creating an ongoing threat that current solutions – such as hired contractors or visual deterrents – have not sustainably addressed. To support the farm in developing a more effective, low-cost deterrent system, our team conducted secondary research focused on three key areas: the nature and risks of avian influenza, the operational context of Wagner Farm, and a range of deterrent strategies – including auditory, visual, and habitat-based solutions. These insights will shape our future design work as we pursue a safe and practical solution tailored to Wagner Farm’s needs.

Avian Influenza

Highly Pathogenic Avian Influenza (HPAI) is a contagious viral disease that poses a serious risk to both domestic and wild bird populations, particularly poultry such as chickens and turkeys. Caused by strains of the influenza A virus, HPAI spreads rapidly and can result in severe illness or death in infected birds. One of the primary transmission vectors is fecal matter from migratory birds, such as geese and ducks, which can contaminate water sources, soil, and pastures. In recent years, the United States has experienced recurring outbreaks of HPAI, prompting major concerns in agricultural and recreational settings where animals, staff, and the public interact [1], [2]. At Historic Wagner Farm, the uptick in HPAI has intensified concerns about the safety of their pasture-raised livestock, the well-being of staff, and the health of visitors.

Currently, Wagner Farm has implemented foundational biosecurity protocols to reduce direct contact between wild birds and their chickens and cattle. However, geese continue to land and gather in open pasture areas, creating a high risk for viral transmission through droppings. The virus can remain infectious in organic material for extended periods, meaning even brief goose visits can contaminate critical spaces [3]. The farm has previously hired external contractors to harass or chase away geese, but these services are costly and not always effective. Moreover, geese are highly adaptive and often become habituated to visual or auditory deterrents, reducing their long-term success.

Through initial information provided to us, it became clear that there is an urgent need for a new, cost-effective, and easy-to-use system that empowers staff to deter geese before they pose a risk. The solution must be compatible with daily workflows, safe for livestock and humans, and humane toward wildlife. Our team focused on designing a proactive goose deterrence system that integrates seamlessly into the farm's environment and operations.

About the farm

The Historic Wagner Farm is a publicly owned farm established in 2001 spanning 18.6 acres that aims to provide an educational and recreational space for visitors to learn more about life on a farm and gain further understanding of local agriculture [4]. The facility conducts a variety of programs catering to scouts, schools, and the general public, and they are open to birthday parties and have a Heritage Center that can be rented. They also facilitate events such as a livestock fair happening in the summer and a farmer's market. The heritage center is open to the public from 9 AM to 5 PM and the farm premises are accessible from dawn to dusk throughout the year. In a typical year, they have about 140,000 visitors [4]. The Friends of Wagner Farm, which is an association that funds the farm, has recently renovated the farm space to create a playground and a garden to enhance the educational component of the farm experience [5].

Visitors are generally not allowed to feed or touch the animals present on the farm. If they are to interact with the animals, visitors must register for farm-sanctioned events specifically catered towards interacting with the farm animals. No dogs are allowed on the premises except for service animals.

Potential solutions

To reduce geese presence and mitigate the risk of avian influenza at Wagner Farm, we researched a range of deterrent strategies. These fall into the following main categories: auditory deterrents, visual deterrents, habitat modifications, and physical deterrents. Each of these categories offers a unique approach with distinct advantages and limitations, especially when considering the open layout and frequency of visitors at the farm.

Auditory Deterrents

Auditory deterrents rely on loud, unexpected sounds to frighten geese and disrupt their sense of safety. Common methods include propane cannons and recording of bird predatory calls [3]. These tools can be effective in the short term, especially in wide-open fields where sound carries well. However, geese can become habituated to repetitive noises overtime, especially if they learn there is no real threat. At Wagner Farm, where public presence is common, loud sound might disturb animals, guests, and staff. Therefore, auditory deterrents may be more appropriate for early morning use or farther from visitor areas.

Visual Deterrents

Visual deterrents work by triggering the geese's natural fear responses to potential predators or unfamiliar movement. Predator decoys such as coyote cutouts (see Fig. 1) are most effective when periodically moved to maintain the illusion of life. Reflective materials and motion-activated sprinkler systems will also be very useful tools in keeping geese away from large areas [4]. These methods are generally low-cost and safe for public settings, making them a viable option for Wagner Farm. However, the geese may adapt if these tools are used without variation.

Habitat Modification

Geese prefer short, open grass that allows them to detect predators easily. By altering the landscape to include taller grasses, shrubs, or patches of denser vegetation, the habitat becomes less appealing to geese. Additionally, ridding fields of potential food sources like spilled birdseed or unsecured trash will help to discourage geese from gathering on the property [4]. This strategy is low maintenance once implemented and supports a long-term reduction in geese presence. At Wagner Farm, these modifications could be integrated into the existing landscape in a way that also supports biodiversity. However, changes in vegetation may take time to grow.

Physical Deterrents

Physical deterrents provide a direct way to prevent geese from accessing sensitive areas. Low fencing and wire grids are effective at restricting movement, especially near water sources or where livestock gather. Overhead deterrents – such a wire – can also prevent geese from landing in specific areas [6]. At Wagner Farm, these barriers can be placed strategically to protect high risk zones like shared feeding areas. While physical deterrents are durable and not prone to habituation, they must be designed to not obstruct farm operations, animal movement, or visitor engagement. Initial installment may be more costly, but the long-term benefit makes it a valuable preventative measure.

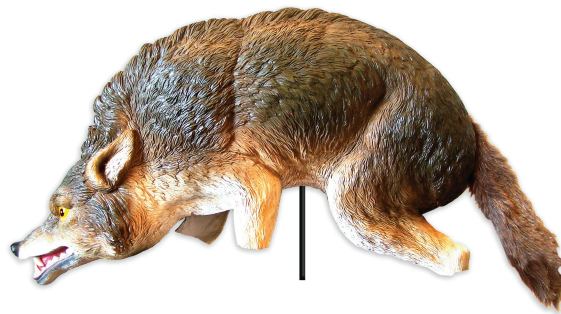


Fig. A1. A potential decoy predator

Conclusion

Understanding how migratory birds contribute to the spread of Avian Influenza is a critical first step in developing strategies to safeguard pasture-raised livestock. Through this secondary research, a range of deterrent methods—ranging from physical barriers and visual scare devices to land management techniques—have been identified and evaluated. The findings suggest that a combination of approaches may be the most effective in minimizing livestock interactions with birds while maintaining the integrity of pasture based farming. These insights will guide our future design work as we develop a practical, sustainable solution to reduce the risk of Avian Influenza on pasture farms.

References

- [1] Centers for Disease Control and Prevention, “Avian Influenza in Birds,” CDC, Feb. 15, 2024. [Online]. Available: <https://www.cdc.gov/flu/avianflu/avian-in-birds.htm>
- [2] U.S. Department of Agriculture, “2022 Highly Pathogenic Avian Influenza (HPAI) Outbreaks,” USDA APHIS, 2023. [Online]. Available: <https://www.aphis.usda.gov>
- [3] World Organisation for Animal Health, “Avian Influenza (Bird Flu),” WOA, 2023. [Online]. Available: <https://www.woah.org/en/disease/avian-influenza/>
- [4] Historic Wagner Farm | glenviewparkdistrict.org. glenviewparkdistrict.org. <https://glenviewparks.org/facilities/historic-wagner-farm/> (accessed 2025-04-09).
- [5] Farm Adventure Area | Friends of Wagner Farm. friendsofwagnerfarm. <https://www.friendsofwagnerfarm.org/farmadventurearea> (accessed 2025-04-09).
- [6] *Goose deterrents | Goose Control products | Bird B gone.* (n.d.). <https://www.birdbgone.com/bird/goose-control/#:~:text=Sound%20goose%20deterrents%20and%20visual,graze%20and%20raise%20their%20young.>
- [7] Control, A. (2024, September 26). *How to get rid of geese in your yard, pond, and property.* Avian Control. <https://aviancontrolinc.com/how-to-get-ri>

Appendix B Summary of Observations at Wagner Farm and Interviews with project partners and biologist

Introduction

The project partner interview with Jonathan Kuester and farm observation was conducted at the Historic Wagner Farm on Friday, April 11 from 4 to 5 PM. The purpose of the session was to learn more about the issues that the geese bring to the farm, as well as previously tried methods. It's crucial that we are aware of the farm environment, such as where the livestock are located within the farm, where the geese tend to graze, and what the area surrounding the farm looks like, for us to come up with a goose deterrent solution that tailors to our project partners' needs. Additionally, it is essential to get more insight on what methods have been tried to deter the geese for us to narrow down what solutions we can potentially use.

The second section of this appendix is notes based on an interview with Seth Magle, who is a wildlife expert at the Lincoln Park Zoo. We conducted this interview to learn more about geese behaviors and prevention methods taken at the Lincoln Park Zoo.

Visit to Historic Wagner Farm and Project Partner Interview

Methods

Interview Structure

The interview consists of a primary interviewer (Victoria Sefen), a tactical notetaker (Audrey Tomen), and a historical notetaker (Tali Smith). We prepared several main questions in advance to learn more about the geese issue on the farm divided into a few key categories:

- Problem Overview: How many geese do you typically see on the property?
- Previously attempted goose deterrence methods: What have you tried implementing to deter the geese? What worked and what didn't?
- Ideal Outcomes: Is there a bar for the amount of geese you wish to be off the property?
- Geese Behavioral Patterns: What time do the geese show up? When do they leave? Where do the geese typically populate?
- Potential Solutions and Considerations: How have the visitors reacted to the geese?
- Livestock Considerations: What animals are the most at risk to the accumulation of the geese?
- Risks: How have visitors reacted to previously used geese deterrence methods?

Results

Setting Observations

With regards to the setting, the farm is a really open area covering about 20 acres. The area is split into three main regions of concern: where the chickens populate, where the cows gather, and where visitors gather. The chickens are confined to a chicken coop, which can be moved to different areas of the farm based on the farmers' discretion. The cows live in a pasture with a barn attached to it; in the pasture there is a small creek, several trees, and a pile of mud. There are no large bodies of water on the property such as ponds, as the rest of the farm is freshly cut grass. The farm is located in a suburban area; several houses have their backyards directly adjacent to the farm premises, there is a church across the street, and the cow pasture faces a busy road that has traffic lights at the intersection. The farm appears to be the largest area of open space within the few blocks surrounding it.



Fig. B1. A photo of the general area within Wagner Farm.

Interview Results

The discussion within the interview started with an overview of the issue. The main issue that the project partners are concerned with is the spread of avian influenza. They indicated that geese have contributed to the spread of avian influenza in both the cows and chickens that are present in the farm. Although the livestock do not directly interact with the geese, avian influenza can spread through geese droppings that are present in varying areas of the farm, such as the cow pasture and areas where the chicken coop may be moved to. The project partners have also said that the geese are creating a mess in the public areas of the farm such as the walkways and the open grass.

Geese Behaviors and Patterns

We enquired about the specific habits of geese on the farm. They are most active in the spring and fall, and prefer open spaces on the farm with high predator visibility. They have adapted to many of the loud noises and lights from the areas surrounding the farm such as the traffic and the

lights from homes nearby, and are able to identify decoy predators. They are spotted in the morning when the farm workers arrive, indicating that they are either present throughout the nighttime to sleep or land on the grass early in the morning, with heavy implication that they are using the farm as a place to sleep. Our project partners have indicated that if the geese do not find the farm as a good place to sleep, they won't come back during the day. The geese are said to travel in groups of 10 to 15, and our project partners have speculated that when the other geese fly by and see that there are geese gathered on the farm, they will join because they feel that it is a safe space. They have also observed that the geese stay away from shrubs 30 inches or taller.

Previously Attempted Geese Deterrence Methods

Our project partners have previously resorted to several solutions to deter the geese. One of them is a goose chasing company that provides a dog to chase away the geese. The dogs cover 4-5 acres of the farmland. They did not specify the working hours or the breed of the dog, but have said that the dog covered the grassy open area as well as the cow pasture. They mentioned that the dog was effective in chasing the geese away, but the cost to care for the dog was a financial burden. The goose chasing company has also employed a laser that was pointed towards a low height so that the geese do not approach the grass (due to city regulations they cannot point the laser into the air or on any reflective surfaces). This was said to be effective in deterring the geese from the area when the lasers were pointed in a side to side motion for most of the day (even in the daytime), but it costs \$1200 a month to maintain the laser. Beyond the goose deterrence company, they tried installing solar powered lights along the fences and grass area; while this was effective in the nighttime, it also disturbed the neighbors and affected the chickens' sleeping patterns. When asked about the chemical repellents, they mentioned that they have not attempted any chemical repellents due to its potential effect on the visitors, especially with scents that may deter the geese.

Livestock Considerations

We also asked about the livestock conditions on the farm, mainly the chickens and cows. A group of chickens live in a movable net fence that the geese don't enter; however when the fence is moved, it can potentially be moved onto a contaminated area. The chicken have similar vision and hearing ranges to the geese, and not all of the chickens go inside at night. They are also not easily agitated, but the project partners have expressed concern towards disrupting the chicken's living patterns at night because that is when they roost, therefore they need a quiet space with minimal light. On the other hand, the cows are less sensitive to the external environment; they are not bothered by the cars on the road, and they generally graze near the pasture or go back into the barn. The cows stay away from tall shrubs as well, and they cannot see in the red spectrum, which is the color laser that they used in their previous solution.

Risks and Complications

The major risks and complications are related to the birds contracting avian influenza: the project partners indicated that if more than 5 birds die because of the disease, they are mandated to report it. Chickens are more at risk than cows, and cows can more easily be checked for the disease, therefore prevention is more dire for the chickens than for the cows. However, when the cows are contaminated by the avian influenza, it affects their milk supply, which impacts the farm.

Ideal Outcomes

Jonathan Kuester has said that the ideal outcomes of the goose deterrence device is full eradication of geese from the farm property, especially near the livestock. He mentioned that the main issue of deterrence is preventing the geese from landing at night and early in the morning when there are no employees present. The goal is to remove geese before the nesting season of next year, which is in April and May.

Potential Solutions and Considerations

We discussed some potential solutions based on our secondary research and possible considerations based on the farm environment and needs. Our project partners have emphasized the importance of a system that continues day and night to prevent geese from staying in the area. They said that using noise may be a good solution because the farm covers a large area. However, the noise may also affect the livestock depending on the noise's proximity to the animals. The cows tend to move around in the pasture, which provides a challenge, although our project partners said that the cows generally tend to stay within a 30 foot radius of the barn. The chicken coop is generally within a few yards of the pasture and is moved around every so often (frequency was not specified). We could also implement a sprinkler system. Although this can't be used during the day when visitors are there, we could implement this in the cow pasture where it is fenced off and visitors cannot go to. We also discussed a drone flying near the birds when they are in the air, although we also came to a consensus that this would likely conflict with airspace regulations in the state.

Legal and Ethical Restrictions

The main ethical restrictions we discussed with Jonathan Kuester is that we are not allowed to hurt the geese. In addition, the solutions cannot be implemented in the month of June, as they are flightless at this time and they are not allowed to affect the nesting process (therefore the solution has to be effective in the months of April and May when they start having eggs). The solution also has to comply with all laws and regulations within the city and state, such as light and noise pollution and airspace regulations. We also cannot pollute the farm or cause any disruptions to the visitors or the public surrounding the farm (the suburban area and the roads).

Discussion

Through this interview we gained some key insights regarding the state of the farm as well as our project's requirements. We learned that the geese presence peaks in spring and fall, especially in the early morning and nighttime. The geese are deterred by movement, sound, and disruption, but adapt quickly to static or predictable methods. The livestock present at Wagner Farm, especially chickens, are highly vulnerable to avian influenza. In addition, the livestock require certain conditions to be maintained to prevent disruption of living patterns (eg: chickens require quiet and low light at night; cows are more tolerant but can still be affected by disease). The geese present on the farm usually avoid tall shrubs and are drawn to open areas where other geese are already gathered. When implementing our solution, legal constraints prevent harm to geese and restrict actions during nesting season, which is in June.

Comparison with Secondary Research

Our secondary research (Appendix A) emphasizes three primary categories of goose deterrents: auditory, visual, and habitat modification. We found that many of the solutions proposed - such as reflective surfaces, predator decoys, motion-activated sprinklers, and unpredictable noise - were either already attempted or considered by the farm.

- **Auditory deterrents** are effective in short bursts but geese quickly become habituated. This aligns with our findings that noise alone has limited long term success and may disturb livestock or neighbors.
- **Visual deterrents** such as lasers and decoys can work temporarily, but geese adapt quickly. Our data confirms that stationary decoys are ineffective, and even laser systems need constant human input to remain effective.
- **Habitat modification** is a promising strategy, noting that geese avoid tall grass and areas where predators may hide. This insight is consistent with the observations that geese avoid tall shrubs, suggesting this could be a valuable addition to deterrence efforts.

Implications for Design: The solution must function continuously - especially at night and early morning - without disturbing livestock or neighbors. In addition, the deterrent must be dynamic or unpredictable to prevent geese from adapting. The design must comply with legal restrictions and avoid harm to geese or creating of public disturbances, and livestock friendly solutions should be prioritized

Interview with Seth Magle

In addition to the interview and end-user observation with our project partner at the Wagner Farm, we also consulted Seth Magle, who is a biologist at the Lincoln Park Zoo, about their goose deterrence techniques and consulted them about different deterrence techniques available (chemical, light, reflective, noise). The consultation was divided mainly into two parts: a

discussion on goose and wildlife deterrence methods at the Lincoln Park Zoo, and his knowledge of goose deterrence and wildlife management.

Geese and Wildlife Management

We learned from Seth Magle that the geese prefer short-cropped grass, are day-active (diurnal) birds that will settle at night to sleep but graze in different grassy areas during the day to eat. Geese are migratory birds, but due to climate change and possibly other factors, they are no longer migrating south during the winter and tend to stay within the midwest/Chicago area even during the cold months. This means that our solution has to work year-round. Geese do not have a good sense of smell. Geese, especially those who settle in urban and suburban areas, are adapted to the noisy and light-polluted conditions and therefore developing a stimulant-based solution becomes quite difficult relative to geese that settle in more rural areas. He recommended that with sound, we can look into their hearing range and see if there are sounds that are not usually emitted by urban areas that would scare off the geese such as wild predator noises.

Solutions implemented by Lincoln Park Zoo

The Lincoln Park Zoo has not implemented any methods to reduce goose populations post-pandemic, but before the pandemic they would remove eggs from goose nests in the spring (April-May) as a long-term geese reduction strategy. Geese make their nests in the ground and therefore the nests are accessible to staff, who would distract the parent geese away from their nests to take the eggs. This was found to be a more humane alternative compared to killing the geese. With regards to bird behavior and deterrence methods, Seth Magle mentioned that they would hold rock concerts at the zoo (not specifically for deterrence) and measure bird activity during and after the concert to see if loud noises were a long-term deterrence technique; however, it was discovered that although the birds stayed away from the premises during the concert, they would come back afterwards and noise did not have a long-term effect.

Appendix C Project Definition

Project Name: Geese Deterrent Device

Project Partner: Historic Wagner Farm

Team Members: Tali Smith, Audrey Tomen, Victoria Sefen, Mika Kato

Date: 6/9/2025

Mission Statement

Design a cost-effective, non-disruptive goose deterrent system that reduces geese presence in shared farm spaces to help prevent the spread of avian influenza while preserving a safe, welcoming environment.

Deliverables

- Final report (digital copy)
- Slide presentation during finals week
- Final prototype

Design Constraints

- 10 week hard deadline
- All materials and designs must be safe for animals and users
- Must comply with geese harassment and wildlife interaction laws

User and Stakeholders

Primary Users

- Farm Workers: Will need to install and turn on device at nighttime

Stakeholders

- Livestock: Interact with geese and contract avian influenza
- Farm Visitors: Consumers of farm product can contract avian influenza if they drink contaminated milk from the cows

Project Partner Profile

Historic Wagner Farm is a public, educational farm offering animal interactions and farm-fresh products. Due to past avian influenza scares, the farm prioritizes the health of livestock and visitors. An effective deterrent system would reduce disease risk and labor burden while maintaining a peaceful environment for animals and the public.

Illustrative User Scenario

In spring and fall, geese frequently gather in the farm’s short-grass pastures—ideal for nesting. Their seasonal presence creates a sustained risk of disease transmission to cows and chickens. A mobile, weather-resistant deterrent system that breaks patterns of geese habituation is needed to prevent flock formation in these zones, especially overnight when cow–geese interaction is most likely.

1. Every evening around 6:30 PM, a farm worker places the solar-powered deterrent post in the middle of the east cow pasture where geese have been congregating. As the sun sets, the motion sensor activates a randomized flashing strobe and predator-call speaker. The lights and sounds startle any nearby geese attempting to land. Because the lights activate intermittently and with random patterns, the geese do not become accustomed to the device and eventually avoid the pasture altogether.
2. On Monday, geese are spotted in the pasture closest to the visitor center, where cows graze during the day. A worker moves the lightweight, weatherproof device from its post and stakes it into the ground in the new hotspot. Since the device is easily relocatable and resets in under five minutes, the farm can flexibly respond to shifting geese activity. The randomized activation ensures effective harassment without overstimulating animals or neighbors.
3. During peak migration in fall, geese tend to land on the pasture before sunrise. To prevent this, the device is set up the night before in a known geese landing zone. When the geese approach at dawn, the combination of flashing light and sound activates before they touch down. The unpredictability of the deterrent—sometimes just light, sometimes light and sound—makes the area feel unsafe for nesting or gathering. Over several days, the geese change flight paths entirely.

Project Requirements

Table 1. Project Requirements

Needs	Metrics	Units	Ideal Value	Allowable Value
Safe for cows	Presence of stress, injury, or exposure to light	Light wavelength	620 to 750	620 to 750
Safe for cows	Presence of stress, injury, or exposure to sound	Decibel range	60-80	60-90
Easy to maintain	Time to set	Minutes	≤ 5	≤ 10

and deploy	up/reset			
Temperature-resistant	Operational temp range	Degrees Fahrenheit	110	100
Water-resistant	Waterproof rating (IP Standard)	IP Rating	IP65+	IP44
Randomized stimuli to prevent geese habituation	Frequency of change	Minutes	10	20
No sharp edges	Minimum edgeRadius	Inches	≥ 2	≥ 3
No sharp corners	Angle of corner	Degrees	≥ 80	≥ 90
No exposed wires	Presence of exposed electrical components	Binary (Yes/No)	No	No
Easy to maintain	Time between maintenance tasks	Days	>12	> 4
Easy to maintain	Time of required maintenance	Minutes	<30	<20

Appendix D Mockup testing and feedback summary

Introduction

We conducted mockup testing at Wagner Farm on May 14, 2025 at 3:30pm as well as separate on-campus testing sessions the preceding week (with each member testing their mockups individually). The team members present for the testing at Wagner Farm were Tali Smith, Audrey Tomen, and Mika Kato, and our project partners Blake Lanphier and Jonathan Kuester. We presented our mockups and a possible final design concept to our project partners at Wagner Farm to get feedback. We also took a further look at cow pasture to figure out the best location to place our design.

Overall we figured out that the wind chimes did not have a strong enough sound to deter the geese from the pasture, so we wanted to create a design with a louder more unappealing sound for the geese. We also realized that just a steady red light alone would not be enough to deter the geese. Therefore we think that a 360° spinning light will bother the geese enough to keep them away from the area. Lastly, we uncovered that the geese gather the most around the creek in the cow pasture so placing a deterrent by that area will be the most effective to keep away the geese.

Methodology

Chime Time (Noise)

Purpose: Test auditory deterrents' effectiveness and user response.

Materials: Hanging chime (metal) and predator face sticker on backing

On Campus Testing

Steps:

1. At lakefront walk up to geese while simultaneously shaking wind chimes to make as much noise as possible

Farm Testing

Steps:

1. **Introduce the mockup:** "This is a sound-based deterrent. It uses wind-activated noise and visual predator cues to scare birds."
2. Blake went into cow pasture with the wind chimes and walked up to the geese shaking the windchimes.
 - Ask Jon:
 - "Would this annoy neighbors or cattle?"
 - "Where would be the most effective place to install the windchimes?"
3. **Discuss alternatives:** "If we wanted to use this in areas without trees, how would you suggest installing it?"

4. Image of Mockup



Fig F.1. Wind Chimes Image

Light It Up (Torch)

Purpose: Test colored light as a deterrent and geese response

Materials: Poster tube with colored film on top

On Campus Testing

Steps:

1. Take colored film and put it in front of phone flashlight (at nighttime)
2. Walk up to resting geese and shine the light in their general direction.
3. See how many geese walk away based on this light.

Note that because we will be going to Wagner Farm during the daytime, we cannot conduct testing on the farm for this prototype. We will discuss the outcomes of these results to our project partners during farm testing.

Farm Testing

Steps:

5. **Introduce the mockup:** “This is a visual deterrent. It can be used to .”
6. Blake went into cow pasture with the wind chimes and walked up to the geese shaking the windchimes.
 - Ask Jon:
 - “Would this annoy neighbors or cattle?”
 - “Where would be the most effective place to install the windchimes?”
7. **Discuss alternatives:** “If we wanted to use this in areas without trees, how would you suggest installing it?”

Reflective Scarecrow

Purpose: Test goose response to reflective material

Materials: Wooden dowel with CDs attached at the top

On Campus Testing

Steps:

1. Go to lakefront with design
2. Walk up to geese with stick moving it around as to reflect the sunlight off of the CDs

Slow Your Roll (sound and light deterrents)

Purpose: Present idea and get feedback on portions

Materials: Battery, solar panel, 360 spinning light, speaker system

Farm Testing**Steps:**

1. **Introduce the mockup:** “This is a sound and light based deterrent which uses a solar panel in order to power the battery”
2. **Questions Asked:**
 - a. Where do the geese flock the most to in the cow pasture?
 - b. Do you think the geese will habituate to the light?
 - c. Will the light bother the cars driving on the highway and the neighboring houses?
 - d. Will the geese get annoyed by the sound?
 - e. Will the sound be too loud and bother the neighbors and cattle?

Results**Chime Time (Noise)**

Observed: Easy to use. Both on the farm and on campus the geese moved away from the noise but were not scared off completely.

Time user spent: > 1 min

Quote of user’s response: “Good idea however the sound was not loud enough to scare the geese off the pasture and cause them to fly away.

Light It Up

Observed: Geese did not move away from the light, no reaction from geese

Reflective Scarecrow

Observed: Reflection had no impact on the geese; even when approached with reflective CDs within 3 feet, they would not move out of the way. There were about 6 geese present in the premises, and all of them stayed in their grazing positions.

Slow Your Roll (sound and light deterrents)

Overall John and Blake were very intrigued by this design. They liked the idea of the red light as it would not disturb the cows but they did have concerns about the noise.

Noise Concerns

1. Worried that sound will cause the neighbors to worry that there is actually a goose in distress
2. Worried that guests will think there is a predator if predator calls are being played.

Quote of user's response: "The light is a very good idea as it will be seen by the geese but not the cows. Make sure that the light and noise do not disturb the neighbors."

Discussion

After our testing we realized that putting our device on the fence of the cow pasture would not work well. We learned that the geese gather the most around the creek in the cow pasture, and the fence post is too far away to have any impact on the geese. So to resolve this issue we decided that it would be best to attach our device to a pole which we can insert by the creek in the cow pasture.

We also realized that the slight clank of the metal from the windchimes was not enough to scare away the geese. We probably will need something louder and more annoying or distressing for them. Our thought is that by playing a mixture of geese distress calls and predator noises it will be unpredictable and scary so that it prevents the geese from landing in that area. In the future we will need to do more research about the correct decibel of noise as to be loud enough to scare the geese but not too loud as to disrupt the neighboring areas.

The last observation that came from our testing is that the red light from our flashlight is not strong enough to bother the geese. We think that since the light has no motion, it did not bother the geese enough to cause them to flee from the area. Our hope is that by getting a rotation 360° light we will be able to bother the geese enough as to prevent them from landing in the area.

Appendix E Instructions for construction and bill of materials

Introduction

Do geese constantly damage pasture areas and put livestock at risk at Historic Wagner Farm?

The Slow Your Roll device offers a low-maintenance, solar-powered solution that emits deterrent sounds at timed intervals throughout the day. This modular and weather-resistant system helps reduce goose presence by playing a goose distress call through a speaker. These step-by-step instructions will guide you through building, assembling, and installing your Goosebusters system from scratch.

Bill of Materials

Table 2. Bill of Materials used to construct Slow Your Roll

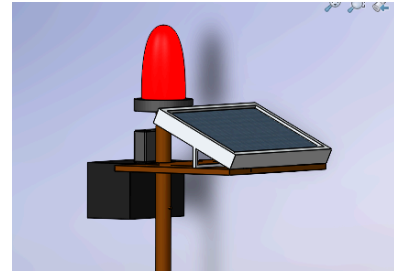
#	Item	Description	Vendor	Qty	Unit Price	Total
1	Bluetooth Speaker	JBL GO2 Waterproof Ultra-Portable Bluetooth Speaker (Black)	Amazon	1	\$31.10	\$31.10
2	Timer Relay Module	UCTRONICS DC 12V Programmable Time Delay Relay with LED Display	Amazon	1	\$12.99	\$12.99
3	Strobe Light	LUBAN 12V DC Red LED Rotating Beacon, No Buzzer	Amazon	1	\$14.99	\$14.99
4	Solar Panel Kit	ECO-WORTHY 10W Solar Panel Kit w/ Smart Controller + Mount	Amazon	1	\$35.89	\$35.89

5	Umbrella Anchor	DAKEPOLE Heavy Duty No-Dig Ground Anchor (Metal, Black, 17")	Amazon	1	\$17.99	\$17.99
6	Battery	12V Sealed Lead Acid Battery (Home Depot)	Home Depot	1	\$17.00	\$17.00
7	MP3 Module	12V USB MP3 Audio Player Module (USB + AUX support)	Amazon	1	~\$7.00	~\$7.00
8	USB Flash Drive	USB Stick (used to store sound file, compatible with MP3 module)	Amazon	1	~\$6.00	~\$6.00
9	12V to 5V Converter	DC-DC Buck Converter 12V to 5V Step-Down Module	Amazon	1	~\$7.00	~\$7.00
10	Waterproof enclosure box	Plastic Waterproof Electronics Project Box	Amazon	1	~\$13.00	~\$13.00
11	Wooden Mount	Constructed Wooden platform to attach to Pole	Constructed	1	~	~
12	Additional Mounting Materials	Mounting brackets, screws, heat shrink tubing, silicone sealant	Variable	~	~	~
Total						\$162.96

Construction Steps

1. Prepare the Post

- a. Cut a 36" length of 2" radius wooden post or pvc pipe using a saw. Sand any rough edges. Drill pilot holes where the solar panel, enclosure box, and speaker will be mounted. Attach the umbrella anchor to the bottom.

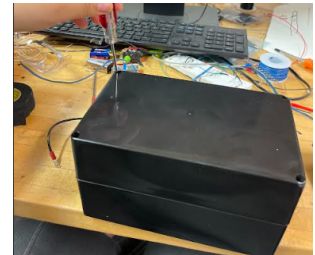


2. Mount the Solar Panel

- a. Secure the 10W solar panel to the top of the post by attaching a wooden mount, and using brackets or screws. Position it to face maximum daily sunlight for optimal charging.

3. Attach the Waterproof Enclosure

- a. Secure the waterproof enclosure midway down the post using Velcro straps or screws. This will house the battery, timer, and MP3 module.



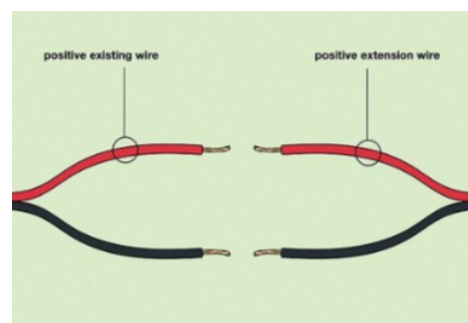
4. Load the Sound File

- a. Download a goose deterrent sound in .mp3 format onto a USB drive or SD card. Make sure the file is named simply (e.g., `goose1.mp3`).



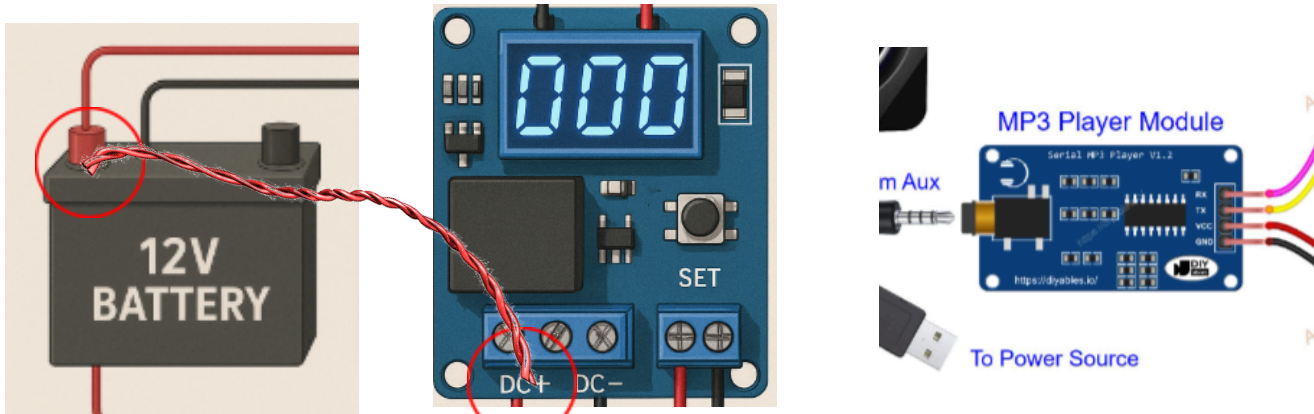
5. Install Electronics

- a. Internal Wiring (General Overview)

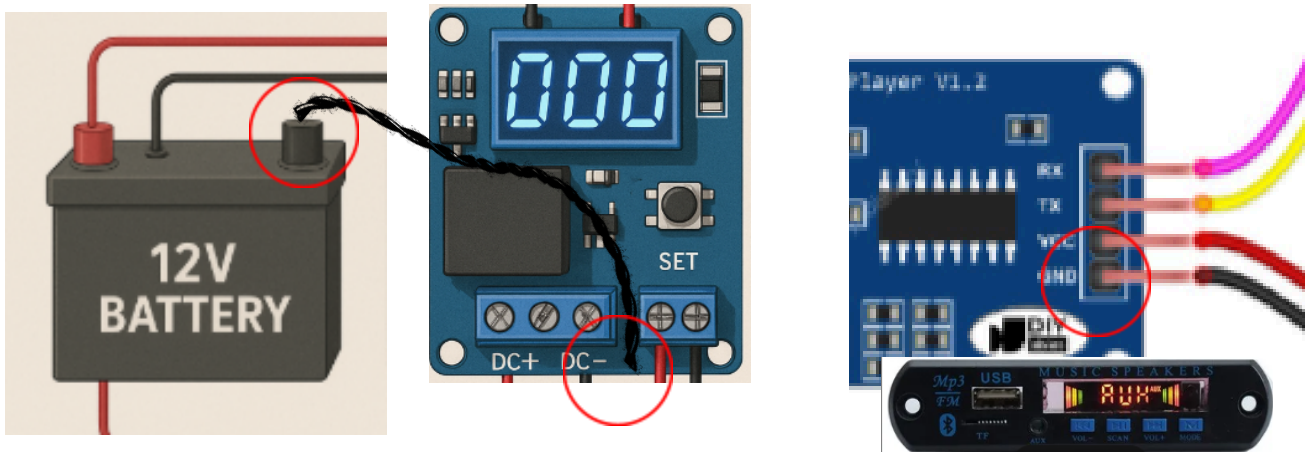


- i. **Before sealing the box, complete and test all electrical connections.**
Identify wires by color: typically, red = positive (+) and black = ground/negative (-).

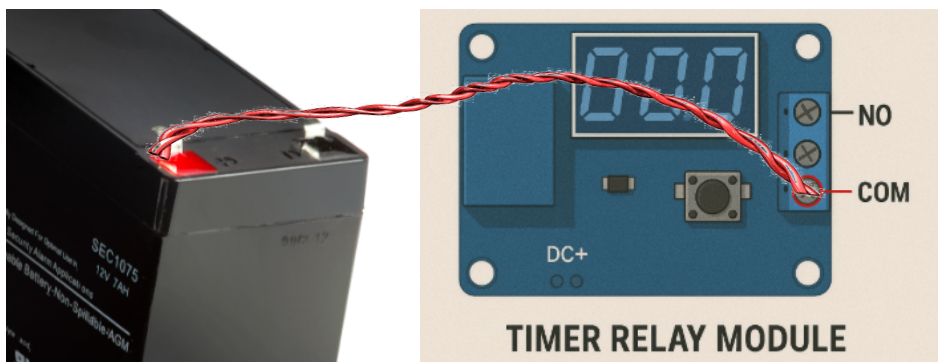
1. Place battery, MP3 module, and timer relay inside the enclosure.
2. Connect the battery's positive (+) terminal to both the DC+ input on the timer relay (*refer to Part 5 section b for further instructions on the timer relay*) and the red power wire on the MP3 module.



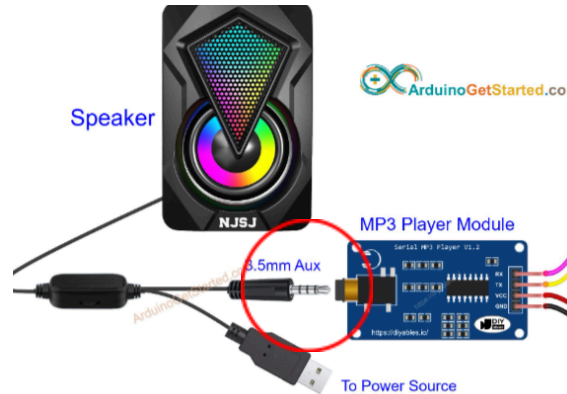
3. Connect the battery's negative (-) terminal to both the DC- input on the timer relay and the black ground wire on the MP3 module.



4. Connect the COM terminal on the timer relay to the battery's positive terminal using a positive red wire.

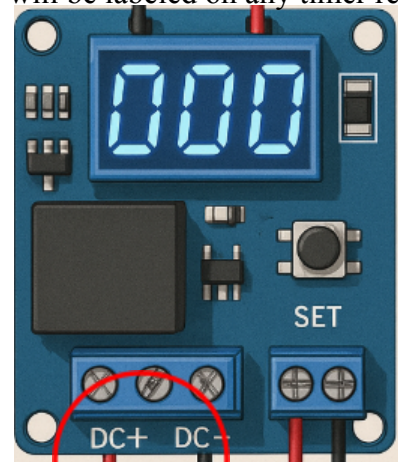


5. Connect the NO (Normally Open) terminal on the timer relay to the positive wire of the strobe light or speaker.
6. Plug a 3.5mm AUX cable from the MP3 module's audio output to the speaker input.



b. Timer Relay Configuration

- i. **The timer relay acts as a switch** that controls when power is delivered to the output device (such as a strobe light or speaker). It receives constant power from the battery but only allows current to flow to the device when the timer is active. To wire it correctly, the battery provides voltage to the relay through the DC+ and DC- inputs, and the COM terminal serves as the power source that the relay switches on or off. When the timer activates, the connection between COM and NO closes, sending power to the connected device. These inputs will be labeled on any timer relay device.



- ii. **The following connections describe how to properly wire a 12V timer relay module** to control power delivery to an output device, such as a strobe light or speaker, using a simple loop (Mode P4) configuration: **(Remember: + refers to + terminal, which we connect to with a + wire, which is red, - refers to - terminal, which we connect to with a - wire, which is black)**

1. **DC+ → Battery +** — Supplies power to the timer's internal



2. **DC- → Battery -** — Completes the power circuit for the timer module.
3. **COM → Battery +** — Serves as the “input” power line the relay will switch to NO or NC.
4. **NO → Light Device +** — Sends power to your device only when the timer is active.

iii. **The timer relay must be configured to automatically turn on and off at regular intervals.** The following steps guide you through setting it to loop mode with a 30-second ON period and a 30-minute OFF period:

1. Hold the **SET** button to enter programming mode.
2. Select **Mode P4** for loop (repeat) functionality.
3. Set **ON time** to **00:00:30** — this means the device will turn on for 30 seconds.
4. Set **OFF time** to **00:15:00** — this means it will remain off for 15 minutes.
5. Confirm and save your settings.

c. **MP3 Module Setup**

- i. **The MP3 module stores and plays the audio file used to deter geese.**

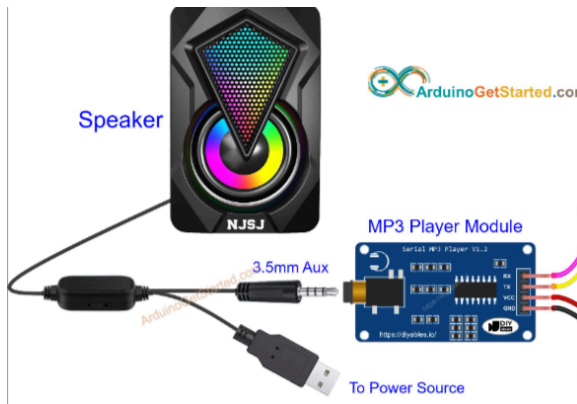
To function correctly, it must be wired directly to the battery for continuous power and connected to the speaker using an AUX cable. The steps below explain how to properly install and activate the MP3 system.

1. Insert a USB stick into the MP3 module containing a single **.mp3** file (e.g., **goose1.mp3**).
2. Connect the red (+) wire from the MP3 module directly to the battery's positive terminal.
3. Connect the black (–) wire from the MP3 module directly to the battery's negative terminal.
4. Plug a 3.5mm AUX cable from the MP3 module's audio output into the speaker's AUX input.
5. After powering on the system, press the onboard play button or use the remote to start audio playback.

d. Speaker Wiring

- i. **The speaker is responsible for playing the deterrent audio stored on the MP3 module.** However, most small waterproof speakers are designed to charge using a standard USB port, which supplies 5 volts—not the 12 volts provided by the system battery. Connecting the speaker directly to 12V could damage it. To solve this, we use a device called a buck converter, which reduces (or “steps down”) the voltage from 12V to 5V. The converter allows you to safely power or charge the speaker using the same battery that runs the rest of the system. The steps below explain how to connect the AUX cable and wire the buck converter correctly.

1. Connect a 3.5mm AUX cable from the MP3 module’s audio output to the speaker’s AUX input.



2. Connect the red wire of the buck converter to the battery’s positive terminal.
3. Connect the black wire of the buck converter to the battery’s negative terminal.
4. Plug the speaker’s USB charging cable into the output port of the buck converter.
5. Turn on the speaker and make sure it is set to AUX mode.

e. Solar Panel Integration

- i. **The solar panel is used to keep the system’s 12V battery charged during the day.** However, to safely manage the flow of electricity between the panel and the battery, you must use a solar charge controller. This device prevents overcharging and protects the battery from voltage spikes. The steps below explain how to wire the solar panel through the controller for safe, continuous charging.

1. Connect the positive wire from the solar panel to the controller’s solar input (+) terminal.

2. Connect the negative wire from the solar panel to the controller's solar input (–) terminal.
3. Connect the controller's battery output (+) terminal to the battery's positive terminal.
4. Connect the controller's battery output (–) terminal to the battery's negative terminal.
5. Check the controller's LED indicators to confirm that charging is active.

f. Final Assembly

- i. **Once all components are connected and tested, you'll need to secure them in place and weatherproof the enclosure.** This step is essential to protect your electronics from movement, moisture, and long-term wear. Follow the steps below to finish the build.

1. Secure the battery, timer relay, and MP3 module inside the enclosure using Velcro, brackets, or zip ties.
2. Bundle all loose wires together and tape them neatly to reduce clutter and avoid strain on connections.
3. Drill small holes in the enclosure for any external wires, such as those going to the solar panel or speaker.
4. Insert rubber grommets into the wire holes to prevent abrasion and water ingress.
5. Confirm the enclosure lid is closed tightly and sealed using built in screws.

g. Testing and Troubleshooting

- i. **Before sealing the enclosure, it's critical to power on the system and confirm that all components function correctly.** This step ensures the speaker, timer relay, MP3 module, and power connections are working before final weatherproofing.
 1. Power on the system using the 12V battery.
 2. Verify that the MP3 module begins playing the audio file from the USB drive.
 3. Confirm the speaker is powered and plays sound clearly.
 4. Wait through a full timer cycle to make sure the light or speaker activates as expected.
 5. If something doesn't work, check the following:
 - a. No sound — Confirm the USB is formatted to FAT32 (*go to the USB tab in your files, right click, select format from the dropdown menu, ensure it is formatted to FAT32*), the file is named simply (e.g., *goose1.mp3*), and that the speaker is powered on and in AUX mode.

- b. Timer failure — Recheck that Mode P3 or P4 is selected, and that COM and NO terminals are correctly wired.
- c. Resets or shutdowns — Inspect all solder joints and verify the battery is charged and delivering stable voltage.

6. Mount the Speaker

Mount the waterproof speaker to the post in a position that allows for clear outward projection. Use screws or mounting brackets to secure the speaker.

7. Deploy the Device

Install the post in the ground at your desired location in the pasture utilizing the umbrella anchor to drive into the ground. Make sure all components are tightly secured, and the solar panel is properly angled.

Appendix F Instructions for use

Listed below are the steps to follow in order to set up Slow Your Roll.

Placement of the pole into the ground

1. Choose a Location

1. Identify a location on the far side of the creek, where geese typically enter the pasture. Ensure the ground is firm and accessible to staff.

2. Install the Base of the Pole

1. Screw the base securely into the soil by hand.
2. Confirm the pole base is level and sturdy.
3. Ensure that the solar panel is facing true south for optimal sunlight capture

3. Insert and Secure the Pole

1. Carefully insert the pole (with mounted system) into the secured ground base.
2. If needed, use additional braces or guy wires to stabilize the pole in windy conditions.

4. Activation

● Daytime charging instructions

1. Ensure the solar panel is securely mounted and positioned for maximum sunlight exposure.
2. Verify that the solar panel's red (positive) and black (negative) wires are connected to the corresponding battery terminals:
 - a. Red wire → Positive (+) terminal
 - b. Black wire → Negative (–) terminal
3. Verify that all other red (positive) and black (negative) wires are disconnected from the corresponding battery terminals:
4. Let the solar panel charge the battery throughout the day.

● Powering On the Device (Evening Setup)

1. Open the enclosure box using a screwdriver.
2. Locate all loose red (positive) wires and attach the looped ends to the positive (+) battery terminal hook.
3. Repeat for all black (negative) wires, connecting them to the negative (–) terminal.
4. Double-check that all connections are secure and match polarity.
5. Close the waterproof box and fasten all screws tightly to reseal the enclosure.
6. The device is now active and will operate based on the timer relay settings.

● Powering Off the Device (Morning Routine)

1. Use a screwdriver to reopen the waterproof enclosure.
2. Carefully remove all red and black device wires from the battery terminals.
3. Leave only the solar panel wires connected to the battery for continued charging.
4. Close and reseal the box, ensuring the solar panel continues charging the battery throughout the day.